Pre-FOMC Announcement Drift After 2020

November 10, 2026

Abstract

We test whether the pre-FOMC announcement drift survives after 2020 and whether it is concentrated in high-uncertainty meetings. Using S&P 500 futures windows ending 15 minutes before the statement, a pre-specified state threshold (VIX 15.07, the development median) and a holdout of 52 scheduled meetings from 2020 to 2026 that was run once, we find a mean excess window return of 38.3 bps, far above matched-VIX non-meeting days (55.5 bps difference, $p = 0.002$). The anxious-state premium of 37.8 bps is not distinguishable from zero with year-clustered wild-bootstrap inference ($p = 0.149$), and the development finding that the premium is larger in anxious states does not replicate (contrast -2.6 bps, $p = 0.946$), but the calm cell has only nine meetings and seven year clusters limit power. A verification of Federal Reserve release times also corrects the development benchmark: eight 2011–2012 meetings were released near 12:30 ET, and with the correct window the development mean falls from 6.2 to 2.8 bps and shows no post-2015 decline. The holdout result is unaffected.

1 Introduction

Lucca and Moench (2015) document large average excess equity returns in the 24 hours before scheduled FOMC policy statements, about 49 bps in 1994–2011, and subsequent work ties the premium to uncertainty (Hu et al., 2022; Martello, 2018). Three questions follow. Does the premium survive publication and the shift to press conferences at every meeting after 2019? Is it concentrated in high-uncertainty meetings, as the uncertainty-pricing account of Hu et al. (2022) predicts? And are the original findings robust to how the announcement time is measured? A premium that shrinks after publication would be consistent with the decay of published predictors (McLean and Pontiff, 2016).

We fix the specification, state threshold and inference procedure in a pre-analysis plan, estimate on 2010–2019 intraday futures data, and run the 2020–2026 holdout once. In the holdout the mean excess window return is 38.3 bps and exceeds that on matched-VIX non-meeting days by 55.5 bps ($p = 0.002$). The anxious-state premium is 37.8 bps but is not distinguishable from zero with year-clustered wild-bootstrap inference ($p = 0.149$), and the anxious-minus-calm contrast is -2.6 bps ($p = 0.946$), against $+19.2$ bps in development. With seven clusters and nine calm meetings the holdout has limited power, so we report bounds rather than evidence of absence.

A second contribution is measurement. Checking each development meeting against Federal Reserve release pages, we find that eight 2011–2012 meetings with press conferences released the statement at 12:20–12:35 ET rather than 14:15 ET, so the frozen window end followed the announcement for those meetings. We correct the development benchmark in a documented post-freeze amendment. The corrected development mean is 2.8 bps rather than 6.2, and the 2010–2014 versus 2015–2019 decline disappears. The holdout, whose statements are released at 14:00 ET, is unaffected. The paper is a disciplined out-of-sample check, not a causal estimate.

2 Related literature

Lucca and Moench (2015) report large average excess equity returns before scheduled FOMC announcements: about 49 bps over the 24 hours ending roughly 15 minutes before the announcement, across 131 scheduled meetings from September 1994 to March 2011 (as stated in the New York Fed staff report version). They find similar pre-announcement returns in other major equity indices but not in Treasuries or money market futures, and no comparable returns before other macroeconomic announcements. The effect is larger when the yield curve is flat, when the VIX is high and when past pre-FOMC returns were high.

Hu et al. (2022) study overnight returns before nonfarm payrolls, ISM and GDP releases and propose that heightened uncertainty about the impact of upcoming news, proxied by the VIX, is priced and then resolves at the announcement. Martello (2018), a 2018 master's dissertation at PUC-Rio (SSRN 3286745), studies state dependence of the FOMC premium on implied volatility: its introduction reports an average drift of 109 bps when implied volatility is above its prior median and 9.7 bps below, in a baseline sample of 60 scheduled meetings from September 2003 to March 2011. This is the closest antecedent to our state split, which is why we fix the threshold from the development sample rather than from the holdout.

A separate strand asks whether the drift reflects information leakage or monetary policy communication. Cieslak et al. (2019) show that equity returns since 1994 concentrate in even weeks of the FOMC cycle and link this to informal communication between Fed officials and market participants. Bernile et al. (2016) find abnormal E-mini S&P 500 futures trading during FOMC news embargoes that predicts later policy surprises. Kurov et al. (2019) find signs of informed trading before about nine of twenty market-moving announcements. If such channels operate, measured premia depend on communication and compliance regimes, which changed with the move to a press conference after every meeting from 2019 (Boguth et al., 2019). That paper reports that announcements with a press conference move prices more and that the market risk premium is higher on press conference days, which matters here because every 2020–2026 meeting had one.

McLean and Pontiff (2016) study 97 cross-sectional predictors and report portfolio returns 26% lower out of sample and 58% lower post-publication. They study cross-sectional predictors, not announcement-window returns, so we use them only as a benchmark for expected decay. Our holdout tests the state-dependent prediction on data that none of these papers used.

3 Data

Prices. The outcome is built from S&P 500 e-mini futures last-trade prices on a fixed Eastern-time grid (09:45, 12:00, 13:30, 13:45, 14:00, 14:15, 14:30, 15:45, 16:00), obtained from Databento. SPY from WRDS TAQ cross-checks the grid on 2010–2017 development days (correlation 0.997 of 24-hour log returns, differences explained by ex-dividend days and one bad SPY print). Contract rolls are flagged and excluded. Futures trade nearly 24 hours, so the window return is measured from the 16:00 cash close of the previous valid session. The outcome is the log return from that print to the print at the window end on the meeting day, net of the daily risk-free rate, in basis points.

Window end. The window ends 15 minutes before the statement. From 2013-03-01 the statement is released at 14:00 ET and the end is 13:45; before that, statements were released at 14:15 and the end is 14:00. Per-meeting release times were checked against Federal Reserve pages for the meetings where the rule could fail: eight 2011–2012 press-conference meetings released the statement at 12:20–12:35 ET (listed in the replication data), and for these the window ends at

12:00. The holdout uses 13:45 throughout.

State and samples. The state variable is the VIX close five trading days before the meeting. The development sample covers 77 scheduled meetings from 2010-06 to 2019 and 2,239 non-meeting days; the holdout covers 2020–2026 with 53 scheduled meetings, of which 52 pass the validity filter. Unscheduled meetings are excluded. Half-day sessions, days with missing trades and futures roll windows are flagged and excluded by the panel builder, which logs row counts at every merge. The anxious threshold is the development-sample median VIX, 15.07; since it depends only on the VIX, it is unchanged by the release-time correction. Intraday data before 2010 are unavailable to us, so the original 1994–2011 evidence enters as published results, not as an estimate. A press conference was held at every 2019 and 2020 meeting, so the press-conference regime is constant in the holdout.

4 Empirical Design

4.1 Identification

The object of interest is the return earned in the window that ends before the FOMC statement is released, in excess of the risk-free rate. Close-to-close daily returns include the announcement reaction and therefore measure a different quantity, so the primary test uses intraday prices. The comparison of interest is between meetings held in calm and in anxious states, where the state is the VIX close five trading days before the meeting (lagged one week) and the anxious threshold is fixed at the development-sample median of 15.07 VIX points. The threshold is not re-estimated on the holdout. The secondary specification replaces the indicator with the continuous lagged VIX.

Identification rests on the timing of the window and on the state being predetermined. Lagging the state by a week limits contamination from anticipation of the meeting itself. The claim is a conditional association between pre-announcement returns and the prior volatility state, not a causal effect of the Committee's actions, and the paper is framed accordingly.

4.2 Threats to inference

Episode clustering. Anxious meetings concentrate in a small number of volatility episodes, so the effective number of independent observations is far smaller than the number of meetings. We cluster by calendar year as an ex-ante proxy for episodes, use CR1 cluster-robust standard errors, and treat the wild cluster bootstrap-t as primary because the holdout has few clusters. Asymptotic t statistics are descriptive only.

Power. With a small number of independent episodes the design can detect a large premium but cannot distinguish a half-sized premium from zero. We therefore report minimum detectable effects from a clustered power simulation using realized volatility, and present non-rejections as bounds on the premium rather than as evidence of absence.

Release-time measurement. The frozen specification assigned release times by rule (14:15 ET before 2013-03-01, 14:00 ET after). Checking each press-conference meeting against Federal Reserve release pages after the holdout run showed that eight development meetings (2011-04-27, 2011-06-22, 2011-11-02, 2012-01-25, 2012-04-25, 2012-06-20, 2012-09-13, 2012-12-12) released the statement at 12:20–12:35 ET, so their frozen window end followed the announcement. The holdout (2020 onward) is unaffected because its statements are released at 14:00 ET. The development

benchmark is re-estimated with those windows ending at 12:00 ET in a documented amendment (Section 5.1); the frozen plan itself is not edited.

Non-FOMC confounds. A premium measured on FOMC days could reflect generic high-volatility or pre-event compensation. The matched-VIX placebo compares FOMC windows with non-FOMC windows on days with similar lagged VIX, so a Fed-specific premium requires an excess over matched days.

State measurement. VIX comparability has changed over time as short-dated option volume has grown. This matters mainly for the exploratory VIX-change test, which is outside the frozen specification.

Specification search. The specification, threshold, outcome window and inference procedure are fixed in the pre-analysis plan before any estimate on post-2019 data, and the holdout is run once. Tests outside the plan are labelled exploratory and cannot change the primary conclusion.

5 Results

All holdout estimates come from a single run of the frozen specification; secondary tests are labelled exploratory. The development column uses the corrected release times (Section 5.1). Standard errors are clustered by year (seven clusters), so we report wild cluster bootstrap p-values alongside.

Primary holdout test. Table 1 reports the primary specification. The anxious-state premium is 37.8 bps ($t = 1.91$, wild $p = 0.149$) against 12.6 bps in development; its sign matches development but it does not reject zero at 5%. The key contrast b (anxious minus calm) is -2.6 bps (wild $p = 0.946$): it does not replicate the development difference of 19.2 bps (wild $p = 0.057$). The reason is the calm cell, whose mean rises from -6.6 to 40.4 bps. That cell has only nine meetings, so b is weakly identified (Figures 1 and 2). The continuous VIX slope is 5.02 bps per point ($t = 2.48$) against 1.87 in development, but it is a single-sample, seven-cluster estimate and two high-VIX meetings in 2020 are visible leverage points in Figure 1. The matched-VIX placebo on non-meeting days gives a Fed-minus-matched difference of 55.5 bps ($p = 0.002$) for all meetings and 38.3 bps ($p = 0.030$) for anxious meetings, so the window return around meetings exceeds that on comparable non-meeting days. In development the same placebo shows no Fed-specific premium (-11.9 bps, $p = 0.65$).

Exploratory secondary tests. The VIX-change test finds no effect: the slope on the five-day VIX change is -3.39 ($t = -0.47$, wild $p = 0.876$). The press-conference subsample (45 meetings from 2021 on) gives an anxious premium of 28.3 bps (wild $p = 0.221$). The Federal Reserve held a press conference at every 2020 meeting too, which we verified only after the single run, so the subsample excludes 2020 for that reason; since the indicator equals one for all 52 holdout meetings, the subsample test adds nothing beyond the primary result and no comparison with non-press meetings is possible. Table 2 shows the anxious premium across thresholds, leave-one-year-out samples, winsorizing and no risk-free adjustment. It ranges from 26.6 to 69.5 bps. No wild p-value falls below 0.05, and the smallest is 0.092.

Table 1: Primary specification, development and holdout. Standard errors (year-clustered) in parentheses. Placebo sample sizes shown are for the holdout; development has 77, 38 and 39 meetings.

	Development (corrected)	Holdout
Calm mean a (bps)	−6.6 (3.3)	40.4 (5.4)
Anxious−calm b (bps)	19.2 (8.3)	−2.6 (21.9)
Anxious premium $a + b$ (bps)	12.6 (7.5)	37.8 (19.8)
Wild-bootstrap $p(b = 0)$	0.057	0.946
Wild-bootstrap $p(a + b = 0)$	—	0.149
VIX slope c (bps/pt)	1.87 (1.05)	5.02 (2.02)
Placebo (all, $n = 52$): Fed−matched (bps)	−11.9 ($p = 0.65$)	55.5 ($p = 0.002$)
Placebo (anxious, $n = 43$): Fed−matched (bps)	−12.0 ($p = 0.81$)	38.3 ($p = 0.030$)
Placebo (calm, $n = 9$): Fed−matched (bps)	−10.8 ($p = 0.59$)	35.2 ($p = 0.209$)

Table 2: Robustness of the holdout anxious premium (exploratory).

Specification	n	n_{anx}	Prem. (bps)	s.e.	Wild p
2021 onward	45	37	28.3	20.5	0.221
Baseline (threshold 15.07)	52	43	37.8	19.8	0.149
Threshold 18	52	30	51.6	20.7	0.092
Threshold 20	52	23	59.2	22.3	0.194
Threshold 25	52	9	69.5	30.4	0.51
Holdout-median threshold	52	26	48.7	25.2	0.284
Drop 2020	45	37	28.3	20.5	0.221
Drop 2021	44	35	51.9	17.2	0.114
Drop 2022	44	35	26.6	20.4	0.256
Drop 2023	44	39	41.6	21.7	0.199
Drop 2024	44	39	34.8	21.9	0.251
Drop 2025	44	35	37.7	24.7	0.3
Drop 2026	47	38	43.8	21.6	0.111
Excluding 2020	45	37	28.3	20.5	0.221
Winsorized 5/95	52	43	35.9	16.2	0.121
No risk-free adjustment	52	43	38.8	19.8	0.135

Interpretation. With seven clusters and nine calm meetings, the holdout is under-powered for the state contrast: the 80% power minimum detectable effect for the anxious half is 40.3 bps at the corrected development volatility (21.8 bps for all holdout meetings). The evidence supports a positive meeting-window return, especially against matched non-meeting days. It does not confirm the development finding that the premium is concentrated in anxious states.

5.1 Post-freeze amendment: development benchmark

After the holdout was run, a check of Federal Reserve release pages showed that eight 2011–2012 development meetings released the statement near 12:30 ET (Section 3), so the frozen window end of 14:00 ET followed the announcement. We therefore re-estimated the development benchmark only, with those eight windows ending at 12:00 ET; the frozen scripts, the threshold and the holdout are untouched. The development mean excess window return falls from 6.2 to 2.8 bps (clustered s.e. 4.6), the anxious premium from 19.3 to 12.6 bps, and the anxious-minus-calm contrast from 26.0

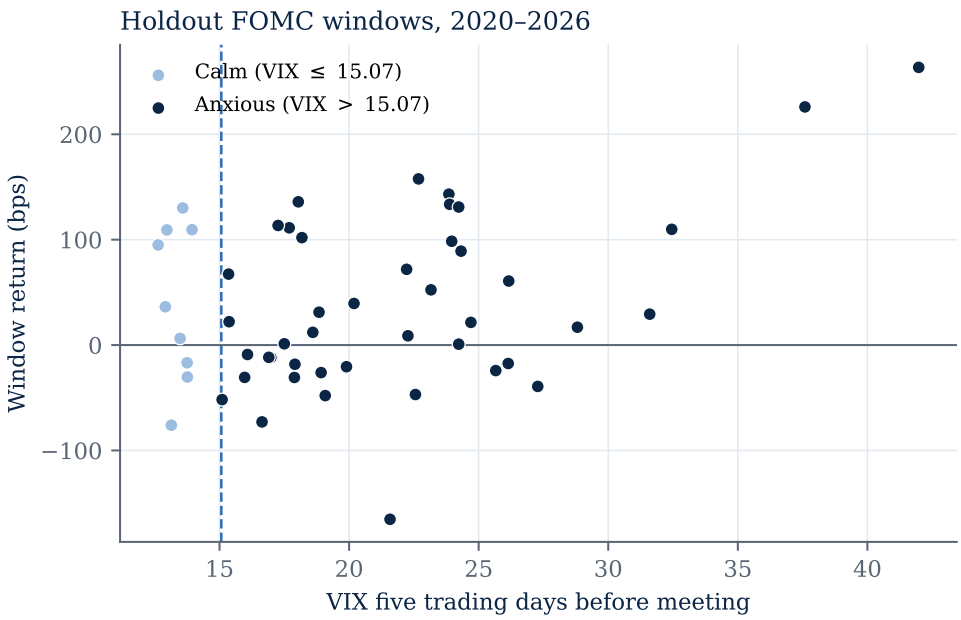


Figure 1: Holdout meeting-window returns against lagged VIX.

to 19.2 bps (wild $p = 0.057$). The 2010–2014 and 2015–2019 means are now 2.9 and 2.8 bps, so the apparent post-2015 decline in the frozen run was produced by the eight mis-windowed meetings. The state contrast keeps its sign under every alternative window end (Appendix A.5). Because the correction was made after seeing the holdout, we flag it as an amendment, not as part of the pre-registered plan; it changes what the holdout is compared with, not the holdout estimates.

6 Discussion

The holdout supports a positive meeting-window return, particularly relative to matched non-meeting days, but does not replicate the development finding that the premium is concentrated in anxious states. Three readings are consistent with the data. First, the premium may be roughly state-independent after 2020; the calm cell mean rose from −6.6 to 40.4 bps and the overall mean from 2.8 to 38.3 bps. Second, the state contrast may be present but undetectable: the calm cell has nine meetings and the minimum detectable effect for the anxious half is 40.3 bps. Third, the 2020–2022 episode, in which VIX was high and the communication regime changed, may dominate a seven-cluster sample. The data cannot separate these.

Limitations are as stated in the design: release times are assigned by rule, and for eight 2011–2012 development meetings the rule is wrong (the statement came at about 12:30 ET), which we corrected after the holdout run (Section 5.1), clustering by year is a proxy for volatility episodes, and the VIX-change and press-conference tests are exploratory. The comparison with the post-publication decline in McLean and Pontiff (2016) is loose: the holdout premium does not fall, but an announcement-window return is not a cross-sectional anomaly portfolio, and the sample is too small to measure a decline precisely.

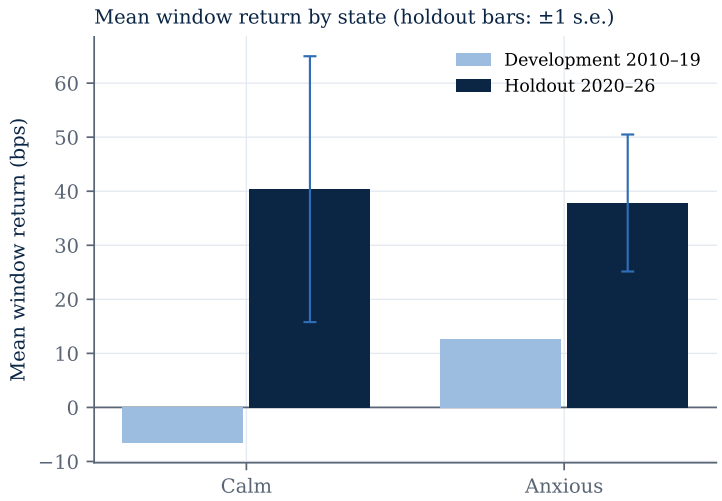


Figure 2: Mean window return by state, development versus holdout.

7 Conclusion

A single pre-specified holdout run finds a mean pre-announcement excess return of 38.3 bps in 2020–2026 that exceeds matched non-meeting days by 55.5 bps, but it does not confirm that the premium is concentrated in high-uncertainty meetings: the anxious-minus-calm contrast is −2.6 bps against +19.2 bps in development. Power is limited by seven year clusters and a calm cell of nine meetings. Verifying release times against primary sources also corrected the development benchmark, which now shows a small premium (2.8 bps) and no post-2015 decline. Natural next steps are per-meeting release times for the whole 1994–2019 history, intraday data before 2010, and a larger post-2020 sample as meetings accumulate.

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A Notation and Additional Material

A.1 Notation

All returns are in basis points unless stated. Meeting m indexes scheduled FOMC meetings; $t(m)$ is the meeting date and $t(m)^-$ the previous valid session.

Symbol	Definition	Unit
P_{prev}	16:00 ET last price on the previous valid session	index points
P_{end}	Last price at the end of the pre-announcement window: 13:45 ET (14:00 ET release, from 2013-03-01 and throughout the holdout) or 14:00 ET (14:15 ET release, earlier)	index points
rf_m	Risk-free rate over the same interval (daily file)	bps
r_m	Outcome: $10^4 \left[\ln(P_{\text{end}}/P_{\text{prev}})\right] - rf_m$	bps
VIX_m	VIX close five trading days before the meeting date	VIX points
τ	Anxiety threshold, fixed at 15.07 (development-sample median)	VIX points
A_m	$\mathbf{1}\{\text{VIX}_m > \tau\}$, the anxious indicator	0/1
a	Intercept: premium in calm meetings	bps
b	Anxious minus calm difference	bps
$a + b$	Premium in anxious meetings (primary test statistic)	bps
c	Slope of r_m on VIX_m in the secondary specification	bps per point
ε_m	Disturbance	bps
$g(m)$	Cluster (calendar year, a provisional proxy for volatility episodes)	—
G	Number of clusters (about seven in the holdout)	—

Table 3: Variables and parameters.

A.2 Specifications

Primary:

$$r_m = a + b\,A_m + \varepsilon_m, \qquad H_1 : a + b > 0. \tag{1}$$

Secondary, reported alongside and not substituted:

$$r_m = a + c\,\text{VIX}_m + \varepsilon_m. \tag{2}$$

Inference uses CR1 cluster-robust standard errors by $g(m)$, wild cluster bootstrap-t (999 draws, seed 1) for $b = 0$ and $a + b = 0$, and a matched-VIX permutation placebo against non-FOMC days (2000 draws, seed 2). With few clusters the bootstrap p-values are primary and asymptotic t statistics descriptive.

A.3 Hypotheses

H1 $a + b > 0$ in the holdout.

H2 The holdout anxious premium is below the development estimate of 19.3 bps.

H3 $b > 0$.

A.4 Sample definitions

Development sample: scheduled meetings from 2010-06 to 2019-12-31. Holdout: scheduled meetings from 2020-01-01. Unscheduled meetings are excluded. A meeting is usable if the meeting day and previous session are valid (no half-day, missing-trade or cash-closed flag) and r_m and VIX_m are non-missing. Release times are a calendar rule, not observed per meeting (UNVERIFIED).

A.5 Development-sample sensitivity

Entries are anxious-state premia in basis points on the 77 development meetings (threshold 15.07), computed by `code/11_sensitivity_dev.py`; the holdout is not involved. Wild p is for the anxious-minus-calm contrast. The first row is the frozen run; the last row is the corrected benchmark used in the paper.

Variant	n	Anxious	Contrast	Wild p
Frozen run (14:00 / 13:45 end)	77	19.3	26.0	0.059
End 13:45, all meetings	77	20.2	26.8	0.062
End 13:30, all meetings	77	19.2	26.5	0.081
End 12:00, all meetings	77	19.8	22.5	0.070
Drop the 8 midday-release meetings	69	7.5	14.1	0.127
Corrected (12:00 end for the 8)	77	12.6	19.2	0.057

The sign of the contrast is positive in every variant, but its size and significance depend on the treatment of the eight meetings. Dropping them leaves a weak, imprecisely estimated state difference (wild $p = 0.127$), which is the cautious reading of the development evidence.

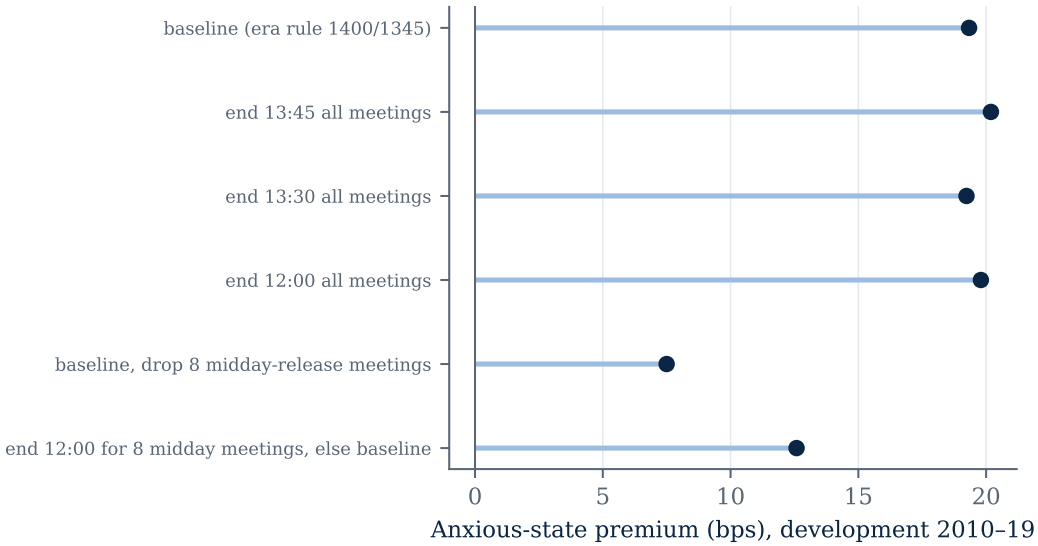


Figure 3: Development anxious-state premium under alternative window ends.